Five Senses Input Simulation Based on Web Resources

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Abstract. Nowadays the technology is still not ready to give a machine the five senses with the recognition level close to the one we use every day and probably will not be ready for a few more decades. But Artificial Intelligence researchers are interested in machine learning based on these senses already today. The cognitive scientists usually develop their theories basing on limited input because gathering the data on what we feel when we touch a stone, smell a perfume, taste an apple, see a thief or hear an explosion would be a very laborious task. Not only inputting such data manually but also gathering those experiences by a robot would be laborious, costly, sometimes not possible at all. In our research we try to make a machine gather the basic knowledge only from the textual level of WWW to make future learning shorter and less laborious as a domain is open. In this paper we will propose a simple method to simulate input from the robot's sensors when only one is really used. By using this method a robot which recognizes an object with one of 5 sensors automatically expects what kind of input would be given by remaining 4 sensors even if they do not physically exist (for example, if the robot using a camera recognizes an object as a stone it automatically should know that normally it has no taste, does not make sound, does not smell and has no taste).

1 Introduction

In our research we are concentrating on the possibilities of automatic commonsensical and emotional information retrieval from the WWW. There are several goals to be achieved with our GENTA project (GENeral belief reTrieving Agent)[1], one of them is to model an average personality based on personal information of hundreds of thousands of Japanese homepages' creators. During our experiments with the *Bacterium Lingualis* method[2] which is used by GENTA for the knowledge and language acquisitions, we noticed the ability of our method to discover nonsenses and to suspect untruth in user's utterances. We also made trials with simple algorithms for robots which usually have no knowledge of what orders are possible for them to perform and which are not

and what kind of tools they need [3]. For instance if a child gives a mischievous command to a robot "bring the floor" and the machine recognizes the object "floor" and the action "to bring", it starts to perform the task not suspecting that user just wants to have some fun seeing the robot scratching the floor before it finds out that the task was not possible to be carried out. Uncomplicated frequency check of "bringing a floor" in the Internet resources, somehow resembles searching our memories for the experiences, or rather lack of experiences saying that somebody somewhere "brought a floor". Lack of some experience alerts our brains and make us suspicious, uncertain, we start to hesitate and reconsider our actions. We claim that before the computers start to seek for the sixth sense or start to understand beauty, nuances and metaphors it should have a solid commonsense built on physical experiences. The question of how the human beings are creating a simple perception from their five senses was a very interesting topic already for our famous ancestors. It was Aristotle [4] who claimed that the commonsense (koine aisthesis) is build upon our five senses and we agree with the Greek philosopher in that point and combine his point with the thoughts of modern thinkers as Minsky[5], Penrose[6], Pinker[7] or Devlin[8]. We agree that the sensors and the fusion of their constant input are the key to providing a machine with an experience. But the technology is not growing that fast and even using a set of the best hardware and software is not able to give us satisfactory results. Hence, the most difficult part of learning by physical experience which is convergence of all sensor inputs into the one interpretation (perception) simply cannot be performed. Our proposition is to use web-mining to simulate the sensor-inputted experiences and to help a machine to create its commonsense based on other people experiences.

The long-distance goal of our research is to reach the state when the machine could simulate input from the 5 senses without using sensors at all. That process could be called *artificial imagination* and it also could be developed further as the machines would be able to use sensors that human beings do not have.

2 Basic methods for simulating experiences

So far we were working on several [1][3][9] topics with one thing in common – we use only Internet resources as they are the biggest and their contents are changing as people's opinions and beliefs. The basic idea is to retrieve needed strings of words, count the frequency and calculate the interdependencies between the comparable items. Some examples will be introduced below.

2.1 The WWW Database

As the Internet is an enormous amount of data, we usually work on parts of it to confirm the correctness of our ideas. We collect only homepages created in Japanese since this language seems to have an easier structure for computer processing especially because of its particles usage what Fillmore has suggested in his works [10]. Another reason are the cultural differences that influence the

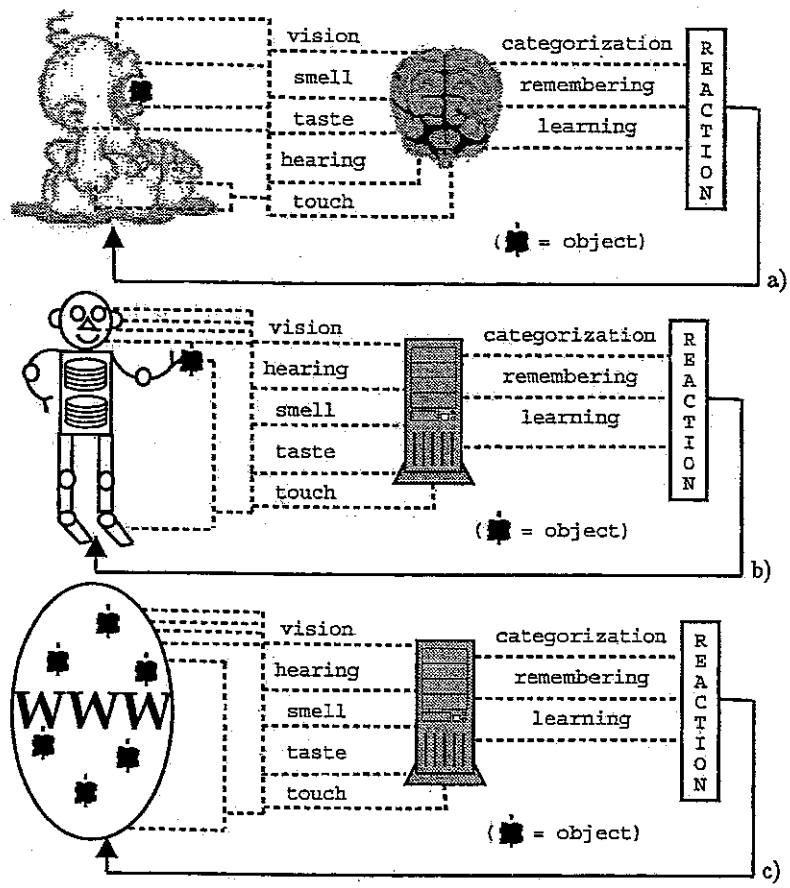


Fig. 1. a) A human child learning from the five sense experiences, b) the ideal robot with perfect sensors and c) the idea of WWW-based supplement for the sensors input memory

commonsense depending on the language, which will be shown later in the paper. For the experiments we have collected almost three millions .jp domain homepages with the Larbin robot, then after filtering off pages without sentences in Japanese and converting them into pure text files, we created a web-based raw corpus consisting of about 2.090.000 documents (approx. 20 Gb). No tagging or whatsoever was conducted. We use Google search engine for manual tests and Google API for semi-manual tests. As it is our first challenge to cope with five senses, most of experiments here were done manually.

2.2 Eyes Input Simulation

For the visual input simulation we chose basic descriptive adjectives (*big, small, high, low, wide, narrow, long, short, bright, dark*) shapes and colors. In every of these three groups a frequency comparison was held to choose three most popular combination and draw a "commonsense border". All combinations beyond this border were treated as "peculiar cases". Fig 2. shows an example where one of the sensors is suggesting an "apple" as a possible object. For instance the tasting sensor gives a signal that the chemical characteristics of the tested object resemble a database entry called "apple". The overall simulation starts and the machine proposes its "5 sense vision" of an average apple, which is done by retrieving "commonsense borders" for other senses. If for example the shape is square, the probability that an object is really an apple becomes smaller.

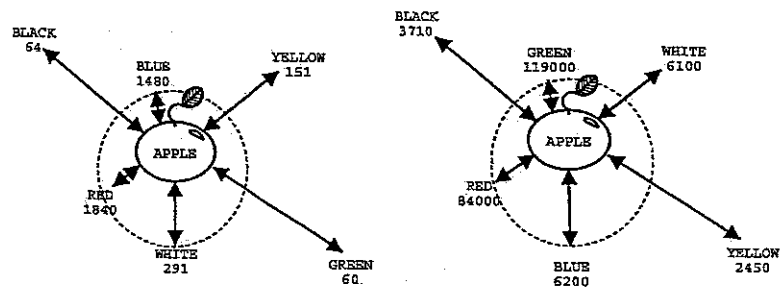


Fig. 2. Comparison between commonsense border for an apple color in case of Japanese (left) and English (right) languages

2.3 Fingers Input Simulation

Perfect haptic sensors are probably the last ones which will be developed. For this input simulation we chose adjectives which are associated with surface characteristics (*hard, soft, wet, dry, smooth, scabrous, slippery, sticky*).

2.4 Tongue and Nose Input Simulations

Probably there is no need to place a tasting sensors in robot's mouth they could be also located in its fingers. But the input should be interpreted the same way as humans do and the recognition output should be one of this category adjectives (*sweet, bitter, sour, salty*). The same situation is with smelling sensors. By using smell-associated expressions as "smells like X", "smells sweet", "smells fishy", etc. we can assume that our robot has smelled every kind of object for several times as there are also flowers that "smell bad". But the simulator suggests that 61 times the smell was good and only 5 times was bad, so *usually* flowers smell nice. The first check is to judge if an object should have any smell at all.

2.5 Ears Input Simulation

The last input is sound sensors input. In GENTA system it also plays a very important role of an user interface when the supervisor is teaching correct answers but in this case where we want to experiment with unsupervised learning, it recognizes only sounds, not a speech. Because the "sound adjectives" as *loud, silent* were not working well in Japanese, we started collecting mimetics (gitaigo) and onomatopoeias (giseigo) to make this input more specific. So far it recognizes only if it is usual for an object to give sound or voices but there are still some results that need deeper analysis as those for a "stone" in a table below.

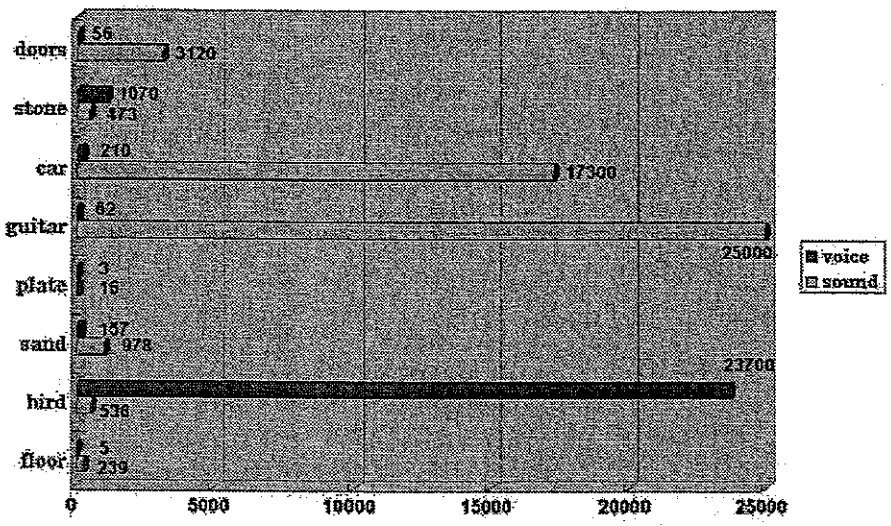


Fig. 3. Frequency check for deciding if an object makes sounds or it give voices

3 Ideas for Sensor Simulation Usage

The main task that our idea could be used for, is abbreviation of the perceptual process. It is much easier for a machine to guess the observed object when more sensors than one are working. However, as we mentioned before, nowadays sensors are far from being perfect. But even for a simple camera it is easier to decide if it is observing a round object than if it is observing an apple. In our previous works that was a human who corrected the computer, but the five senses could be an easy way to make a robot confirm its choices by itself. However, web-mining opens the door for much wider range of applications. Two of them are introduced below.

3.1 Object Abnormality Alert

Let us assume that a robot sees a dark brown object in a basket with something what was recognized as green apples. The input from the visual sensors for a green apple is *round* and *blue* (Japanese mode) while the unknown object has different color - *brown*. Haptic sensors compare two objects and discover a difference in the softness value, the same with the smell input, which is zero value in the case of green apples. Let us say the taste sensor failed to recognize the input. Normally, machines without very complex database, would judge that the object is not an apple. But by using simple web-mining techniques it can be easily confirmed that apples are green and brown, soft and hard, and sometimes they smell. It is also possible in many cases to find the reasons of peculiarity (the brown apple may be rotten, may be baked) by using "if sentences" what was confirmed in our previous work [2]. The same "if sentences" may be used in a slightly different way, which will be described below. Another kind of alerts are the simpler ones given directly to the user which is in danger. If touch sensors detect abnormal temperature of the floor, the system can easily discover that it is not usual (3680 hits of "cold floor" against 41 hits of "hot floor") and warn the user.

3.2 Propositions of Actions

If the system managed to recognize abnormality as a "rotten apple" and the task is for example "bring me an apple", there is a possibility that the robot brings us a rotten apple what is not desired. But we developed a simple method that can help the computer to foresee human reaction in such case [9], which will be described in the next subsection. At first, the machine will know that "rotten apple" has a negative connotation. Then, it will be able retrieve the information about what people usually do with such objects. It is done with similar method as in the case of adjectives but now the verbs are the elements that create a "commonsense border". The methods are identical as in our *Bacterium Lingualis*, the Japanese direct object particle "wo" is added to the object and with the help of Japanese morphological analyzer "Chasen", the most frequent verb retrieval is performed. Then, the system can compare sets of verbs

for an "apple" and a "rotten apple". As we have confirmed, in case of "rotten apple", the number of the verb "to eat" drops drastically and Positiveness of verbs differ in both cases. When it is only an apple we usually eat them, look for them, buy them, but if they are rotten most popular verbs are "put away", "throw up", "throw", etc. Mining with other particles is also performed, it will be described in the subsection about categorization. When we know what is the object usually used for, we can also update the knowledge about the object with the consequences of most common actions toward the object. For that we use three Japanese "if expressions" (V-tara, V-eba, V-to), which has a slightly different semantic connotations. If the state has a low Positiveness value, the system has a will to change this state (about our Good Feeling Pursue Hypothesis, please refer to [2]) and above mentioned method is able to attempt to verify the consequences of chosen action. For instance if a robot for housekeeping has an order to keep the kitchen table clean it must know that trashing a plate just because it is against the "clean state" would not make the user happy. As for such deep mining the Internet resources are still not enough, we are working on combining thesauri and Positveness calculations to develop more sophisticated consequences foreseeing.

3.3 Role of Positiveness

The authors agree with Minsky [11] and Damasio [12] that these are emotions and commonsense which are the key for the future of A.I.. As it was possible to notice above, the emotional information about the objects is a part of most of our methods. The idea of Positiveness can be briefly introduced by the following formula:

$$Positiveness = \frac{C_{\alpha_1} + C_{\alpha_2} * \gamma}{C_{\beta_1} + C_{\beta_2} * \gamma}$$

$$\alpha_1 = disliked, \alpha_2 = hated$$

$$\beta_1 = liked, \beta_2 = loved, \gamma = 1.3$$

Where γ is to strengthen the "love" and "hate" opinions. This method recognizes if an object is **very positive** (Positveness = 5), **positive** (P = 4), **indifferent** (neutral) (P = 3), **negative** (P = 2) or **very negative** (P = 1), and can provide common information about what humans feel toward the given object. For instance, if the object is a "beer" the reaction is positive, when the object consists of two elements: "cold" and "beer", very positive value (P5) is achieved. In the case of an unusual combination of "warm" and "beer", system receives a strongly negative value.

3.4 Categorization

As we were concentrating on developing an universal automatic categorization method for NLP applications, we put stress on particles which are the key for building the categories (nouns that are used most frequently with a given set



Fig. 4. The change of opinion depending on the temperature of beer – the darkest part is the most negative one

of particles becomes one category, the most frequent pairs of object-verbs and adjective-objects creates sub-categories [2]. The reason for such peculiar approach is that we assume these are the particles which stand on the line between the lexics and semantics. Our idea is that if they decide about the categories and subcategories of nouns, there is a greater possibility of simplifying idiom or metaphor understanding. But probably in the case of physical world and robots which are actually touching and smelling things, the senses input could replace particles on the way to achieving a commonsense. The research on sensor fusion [13] is now concentrating on the motion level of a robot, but in the future it may also create another approach to metaphor understanding. During the tests we noticed that there are many adjectives, as for example "soft" (*yawarakai*), which appear in more than one sensor simulator. There is a chance that what is learned by haptic sensors can be used by other sensors. When the system is given a description "soft" to the sound or voice it could use the associations form the haptic input experience to properly interpret its meaning. We think that will be the first place where the learning should be performed.
Our preliminary experiments with "commonsense border" of sensor inputs were strongly influenced by missing adjectives for sound, taste and touch sensors, so we must make computer find them by itself in the future.

4 Conclusions and Future Work

In this paper we introduced several ideas for web-based support for the robots learning by experience. We tried out our NLP methods to support the machines working in the physical world as their number will be growing constantly. Even when the 5 kind of sensors symbolizing 5 senses are able to transmit proper information, there are cases which stand against the commonsense and the appropriate reactions based on the experiences are needed. However, gaining various experiences by a robot or a group of robots in different environments would need a lot of money and time. Therefore we assumed that senses are supplementing each other (the same situation as the spatial knowledge of blind children is growing even without visual input [14][15]) and treating other people experiences as own ones would help to save a lot of money and time. Such system, if realized, could also become an interesting topic for the sociology or psychology researchers as a model of Japanese average "Mr. Internet" was created. The

"average commonsense" could also be a safety valve if we program the machine to make ethical decisions upon the mutual beliefs. Briefly speaking, robot with GENTA inside would not kill anyone just as the *most of us* would not do it. The same idea could be used to develop a robot-supervised learning where a *model* of human is supervising the machine's choices.
Even if the preliminary tests seem to be promising, the ideas given above are only the blocks to be first discussed and thoroughly tested before choosing one of the applications for development. We still have a lot to do for eliminating frequency counting errors coming from even a very small mistakes in regular expressions. We also need to develop the new categorization method and compare it to the previous one which is based on particles.

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